Task 5: Power Budget & Battery Selection

1. Current Draw Estimation (Power Budget Table)

The power budget accounts for all electrical components listed in the project Bill of Materials (BOM). The system operates on a dual-rail power architecture: an **unregulated 7.4 V DC motor rail** directly powered by a 2S LiPo battery, and a **regulated 5.0 V / 3.3 V logic rail** stepped down via an LM2596 buck converter.

Component	**Operating Voltage**	**Idle Current (mA)**	**Nominal Running Current (mA)**	**Peak / Stall Current (mA)**	**Power Rail / Source**	**Rationale & Datasheet Reference**
ESP32-WROOM-32	$3.3V$ (via $5V$ Vin)	80	80	240	Regulated $5V$ Output	$80mA$ active CPU execution; up to $240mA$ peak during Wi-Fi/BT diagnostic transmission.
Pololu QTR-8A Array	$3.3V/5.0V$	100	100	100	Regulated $3.3V$ Output	$8 \times$ IR emitter/receiver pairs drawing up to $100mA$ continuously when enabled.
TB6612FNG Driver (Logic)	$3.3V/5.0V$	2	2	2	Regulated $3.3V$ Output	Quiescent logic supply current (I_{CC}).

GA12-N20 Motors ($\times$ 2)	$6.0V$ $-$ $7.4V$	0	140 ($70mA$ ea.)	2000 ($1000mA$ ea.)	Direct $7.4V$ Battery Rail	$70mA$ rated no-load running current per motor; $1.0A$ stall current per motor.
LM2596 Buck Converter	$7.4V$ Input	10	10	15	Direct $7.4V$ Battery Rail	Internal quiescent operating current (I_Q).
System Total	—	$192mA$	$332mA$	$2357mA$ **(** $2.36A$**)**	$7.4V$ Nominal Pack	Worst-case total includes concurrent dual-motor stall and Wi-Fi burst.

2. Runtime Calculation

Battery Specification**s**

- Battery Selected: Bonka 7.4 V 2200 mAh 25C 2S LiPo Battery
- Nominal Pack Voltage (V_{nom}): $7.4V$
- Battery Capacity (C): $2200mAh$ ($2.2Ah$)
- Derating Factor **(**DF**):** 0.75 (Accounts for thermal losses, voltage drop under load, DC-DC converter inefficiency, and avoiding deep discharge below 20% capacity)

Runtime Formula

$$Runtime(hours) = \frac{BatteryCapacity(mAh)}{AverageSystemCurrent(mA)} \times DeratingFactor(0.75)$$

Case A: Nominal Operating Load (Line Following on Flat Surface)

Under normal track conditions, the ESP32 operates without Wi-Fi enabled ($80mA$), the QTR-8A array is active ($100mA$), and both GA12-N20 motors draw their rated current ($70mA$ each).

$$AverageCurrent(I_{nom}) = 80 + 100 + 2 + 140 + 10 = 332mA$$

$$Runtime_{nom} = \frac{2200mAh}{332mA} \times 0.75 \approx 4.97hours(\approx 298minutes)$$

Case B: Dynamic Racing Load (High-Speed PID Maneuvering & Telemetry)

During real-time line following and maze exploration, system power consumption fluctuates significantly due to continuous differential PWM motor acceleration, sharp turn braking, and optional wireless sensor diagnostics.

Operating Conditions & Assumptions

- Battery: Bonka $7.4V$ $2200mAh$ 2S LiPo Battery ($2200mAh$ capacity).
- Derating Factor (DF): 0.75 (Accounts for thermal losses, voltage sag under dynamic loads, buck converter efficiency, and maintaining a 20% safety margin to prevent LiPo over-discharge).

1. Best-Case Dynamic Scenario (Smooth Traversal & Minimal Telemetry)

- ESP32: Standard active processing without Wi-Fi enabled ($80mA$).
- QTR-8A Array: Continuous IR sensing ($100mA$).
- TB6612FNG Driver (Logic): Quiescent logic supply ($2mA$).
- LM2596 Buck Regulator: Quiescent current ($10mA$).
- Motors ($\times$ 2 GA12-N20): Smooth track continuous drawing $\sim$ $150mA$ per motor ($300mA$ combined).

$$TotalBest - CaseCurrent(I_{best}) = 80 + 100 + 2 + 10 + 300 = 492mA$$

$$Runtime_{best} = \frac{2200mAh}{492mA} \times 0.75 \approx 3.35hours(\approx 201minutes)$$

2. Worst-Case Dynamic Scenario (Aggressive PID Maneuvering & Continuous Wi-Fi Diagnostics)

- ESP32: High-frequency Wi-Fi / Bluetooth telemetry logging active ($240mA$ peak burst).
- QTR-8A Array: Continuous IR sensing ($100mA$).
- TB6612FNG Driver (Logic): Quiescent logic supply ($2mA$).
- LM2596 Buck Regulator: Higher load operating current ($15mA$).
- Motors ($\times$ 2 GA12-N20): Heavy dynamic load during continuous tight-radius turn corrections and rapid direction switches drawing $\sim$ $100mA$ per motor ($2000mA$ combined).

$$TotalWorst - CaseCurrent(I_{worst}) = 240 + 100 + 2 + 15 + 2000 = 2357mA(1.36A)$$

$$Runtime_{worst} = \frac{2200mAh}{2357mA} \times 0.75 \approx 0.7hours(\approx 42minutes)$$

3. Cell Chemistry Justification

For miniature line-following and maze-solving robots, selecting between **Lithium Polymer (LiPo)** and **Lithium-Ion (Li-ion 18650)** involves balancing continuous current capability, total weight, energy density, and physical footprint.

Parameter	**Lithium Polymer (LiPo Pouch - 2S 7.4V**	**Lithium-Ion (Li-ion 18650 - 2S 7.4V**	**Impact on Robot Performance**
C-Rating & Peak Current	**High (**$25C - 50C$ **continuous)**	Low to Moderate ($1C - 3C$ standard cells)	High C-rating prevents logic brownouts/resets during sudden motor stall spikes.
Max Safe Discharge Current	$2.2Ah \times 25C = 55.0A$	$2.2Ah \times 2C = 4.4A$	LiPo easily absorbs the $2.36A$ stall current spike without voltage sag.
Gravimetric Energy Density	High ($\sim 180 - 220Wh/kg$)	**Very High (** $\sim 200 - 260Wh/kg$**)**	Li-ion stores slightly more energy per gram, but requires heavy steel casing.
Weight & Form Factor	**Ultra-Lightweight (** $\sim 110g$ **flat pouch)**	Heavier ($\sim 100g - 110g$ + holders $\approx 135g$)	Lower overall chassis mass allows higher PID acceleration and cornering speeds.
Internal Resistance (IR)	**Very Low (**$< 10m\Omega$**)**	Moderate ($30 - 80m\Omega$)	Low IR ensures minimal voltage drop when high PWM power is applied.

Cost	₹1, 339 (Pack with XT60 & JST-XH)	₹600 − ₹900 (Cells + dual 18650 holder)	Slightly higher cost is offset by performance advantages and ready-made connectors.

Final Justification for LiPo Selection

1. C-Rating & Instantaneous Peak Current: The worst-case stall current of the robot reaches $2.36A$. Standard budget 18650 cells ($1C - 2C$) operate near their maximum threshold during motor stalls, causing severe internal voltage drop that can reset the ESP32 microcontroller. The selected 25C LiPo pack safely delivers up to $55A$ **continuous discharge**, ensuring stable voltage rails under all operating conditions.
2. Weight Optimization for Dynamic Control: In line-following robots, lower mass reduces rotational inertia during sharp turn corrections. The LiPo pouch eliminates heavy cylindrical steel enclosures and battery holders, minimizing overall robot mass for faster response to PID motor commands.